Laboratory Report # 4

Name: *Cyril Andre B. Durango* **Date Completed:** *October 4, 2024*

Laboratory Exercise Title: *Dataflow Modeling of Combinational Circuits*

Target Course Outcomes:

CO1: Create descriptions of digital hardware components such as in combinational and sequential circuits using synthesizable Verilog HDL constructs.

CO2: Verify the functionality of HDL-based components through design verification tools.

Exercise 4A: 4-Bit Comparator. Under a new project, create a Verilog HDL description of a 4-bit comparator using dataflow modeling. Refer to the entity diagram and output details below for reference. Synthesize the design and simulate to verify the results. Show both a simulation log (standard output) and simulation waveform (timing diagram) in the laboratory report.

The 4-bit Comparator Verilog description file utilizes logical & arithmetic operators and ternary operators rather than building the circuit ground up with gate primitives.

Comparator_4bit.v

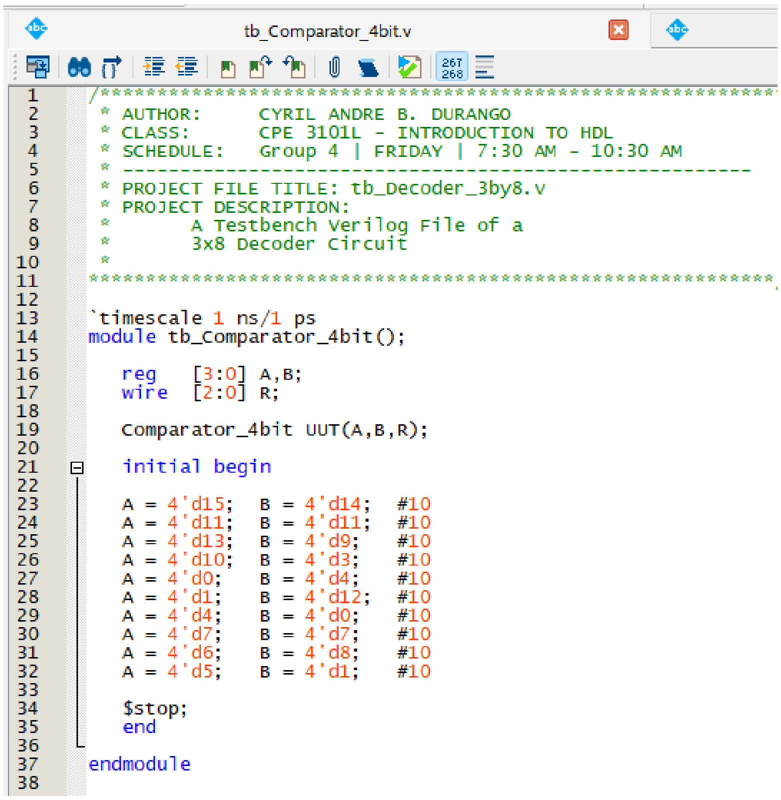
```verilog
/**************************************************************
 * AUTHOR:      CYRIL ANDRE B. DURANGO
 * CLASS:       CPE 3101L - INTRODUCTION TO HDL
 * SCHEDULE:    Group 4 | FRIDAY | 7:30 AM - 10:30 AM
 * -------------------------------------------------------
 * PROJECT FILE TITLE: Comparator_4bit.v
 * PROJECT DESCRIPTION:
 *       A Verilog File of a 4-bit comparator circuit
 *
**************************************************************/

module Comparator_4bit(
   input    [3:0]    A, B,
   output   [2:0]    R
);

   assign   R[2] = ( A > B ) ? 1 : 0;
   assign   R[1] = ( A == B ) ? 1 : 0;
   assign   R[0] = ( A < B ) ? 1 : 0;


endmodule
```

Figure 1. Verilog File of a 4-bit Comparator Circuit



```verilog
/*********************************************************************
 * AUTHOR:      CYRIL ANDRE B. DURANGO
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 * SCHEDULE:    Group 4 | FRIDAY | 7:30 AM - 10:30 AM
 * ------------------------------------------------------------
 * PROJECT FILE TITLE: tb_Decoder_3by8.v
 * PROJECT DESCRIPTION:
 *       A Testbench Verilog File of a
 *       3x8 Decoder Circuit
 *
 *********************************************************************

`timescale 1 ns/1 ps
module tb_Comparator_4bit();

   reg    [3:0] A,B;
   wire   [2:0] R;

   Comparator_4bit UUT(A,B,R);

   initial begin

   A = 4'd15;   B = 4'd14;   #10
   A = 4'd11;   B = 4'd11;   #10
   A = 4'd13;   B = 4'd9;    #10
   A = 4'd10;   B = 4'd3;    #10
   A = 4'd0;    B = 4'd4;    #10
   A = 4'd1;    B = 4'd12;   #10
   A = 4'd4;    B = 4'd0;    #10
   A = 4'd7;    B = 4'd7;    #10
   A = 4'd6;    B = 4'd8;    #10
   A = 4'd5;    B = 4'd1;    #10

   $stop;
   end

endmodule
```

Figure 2. Testbench File of a 4-bit Comparator Circuit

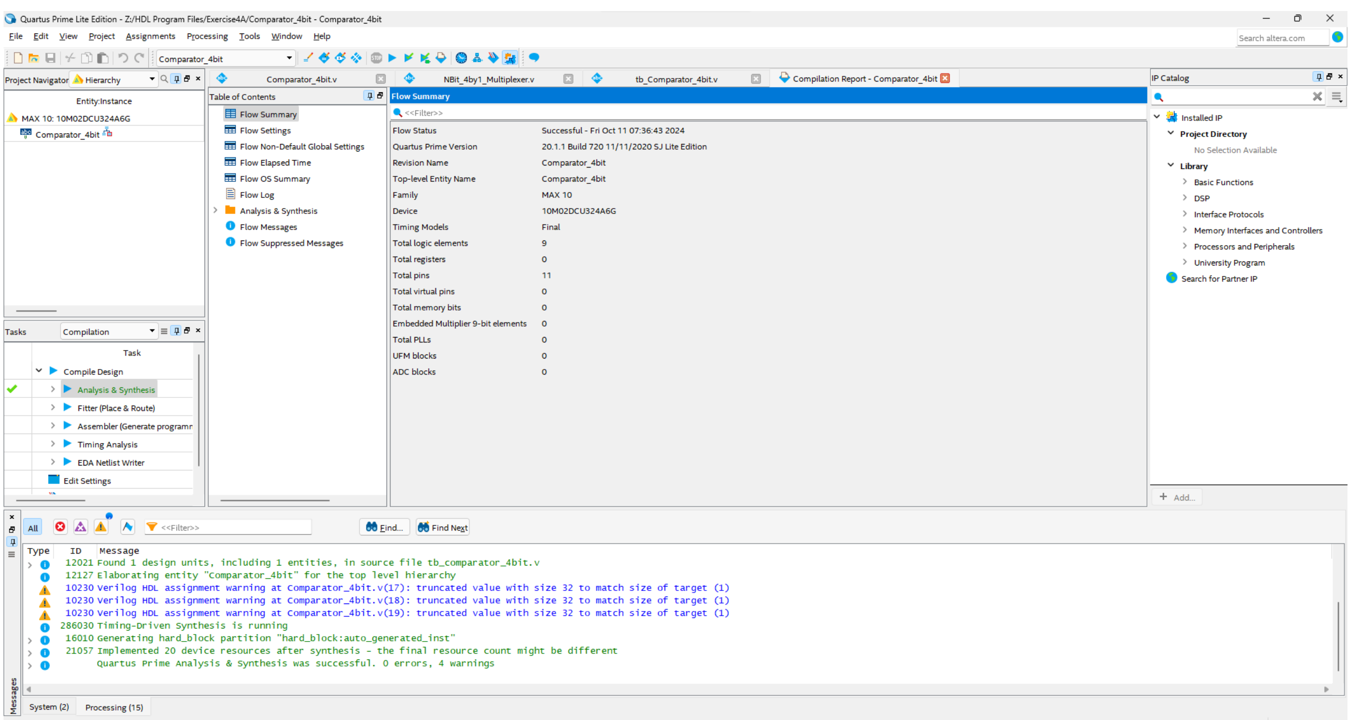
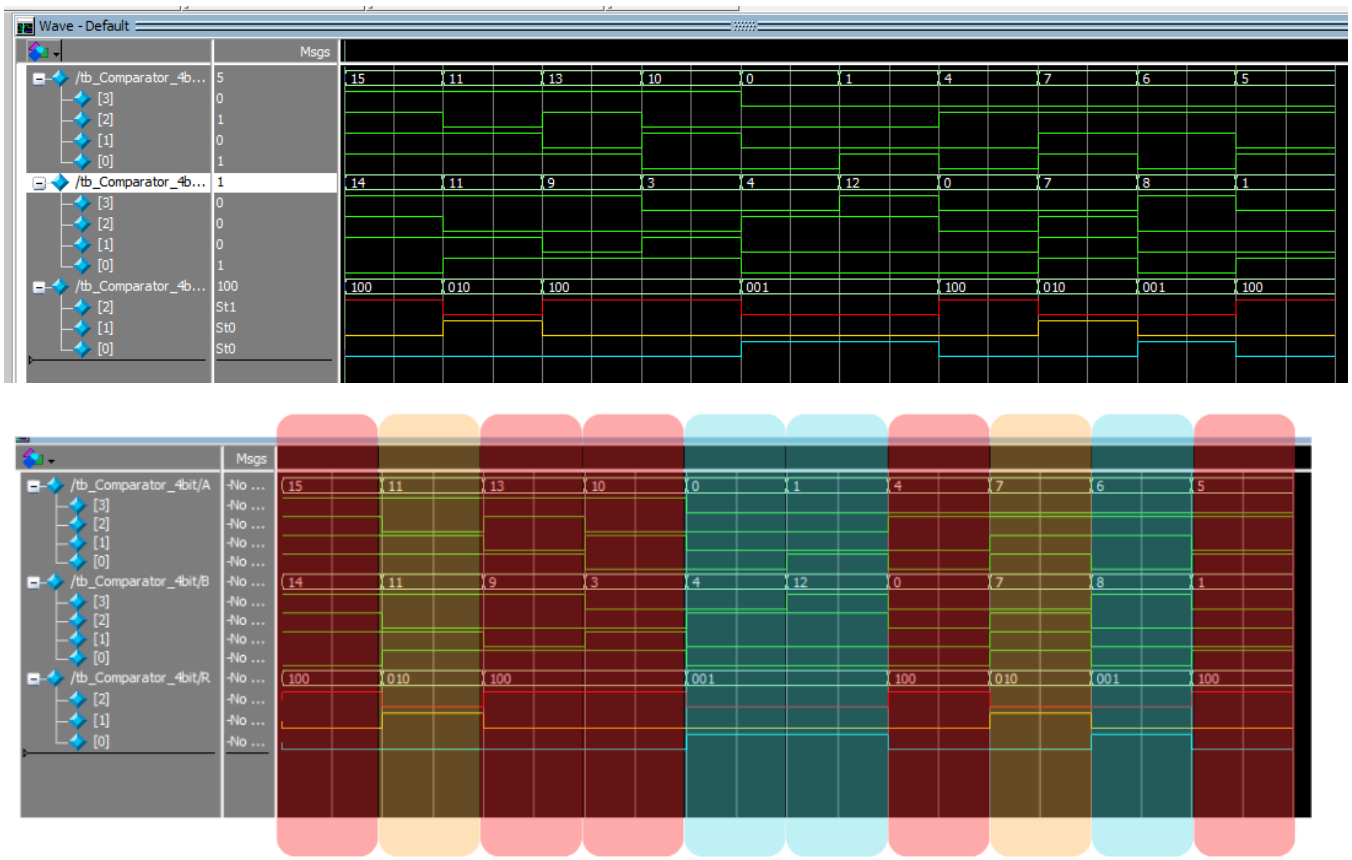


Figure 3. Verified Summary Compilation for the Verilog and Testbench Files of a 4-bit Comparator



red - A > B
yellow - A = B
cyan - A < B

Figure 4. Generated Waveform of the 4-bit Comparator Testbench File, with and without annotations.

The generated waveform demonstrates the logic behind a comparator. In the output waveform, red denotes that A is greater than B, yellow indicates that A is equal to B, and cyan denotes the A is less than B. As seen in the highlighted version, each of the highlighted columns corresponds to a logic-high output in the output waveforms. Those highlighted in red can be seen that A is greater than B, with its corresponding output of 100 that indicating so. The same logic can be applied to the rest of the highlighted areas.

Exercise 4B: n-Bit 4-to-1 Line Multiplexer. Design a dataflow description of an n-bit 4x1 multiplexer. Use a parameter to change the value of n (use n = 4 as the default value). Refer to the entity diagram and truth table below for reference. Synthesize the design.

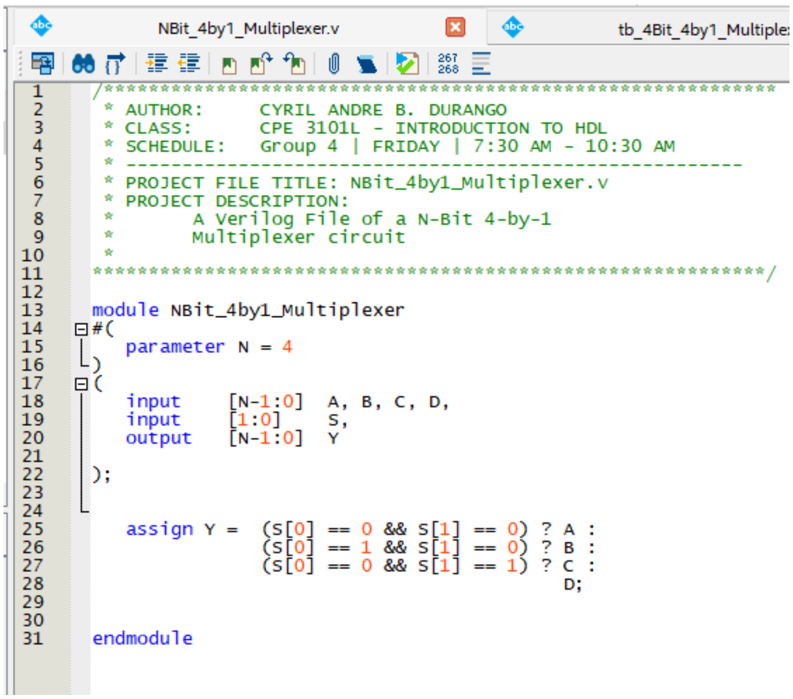


Figure 5. Verilog File of an N-bit 4-by-1 Multiplexer

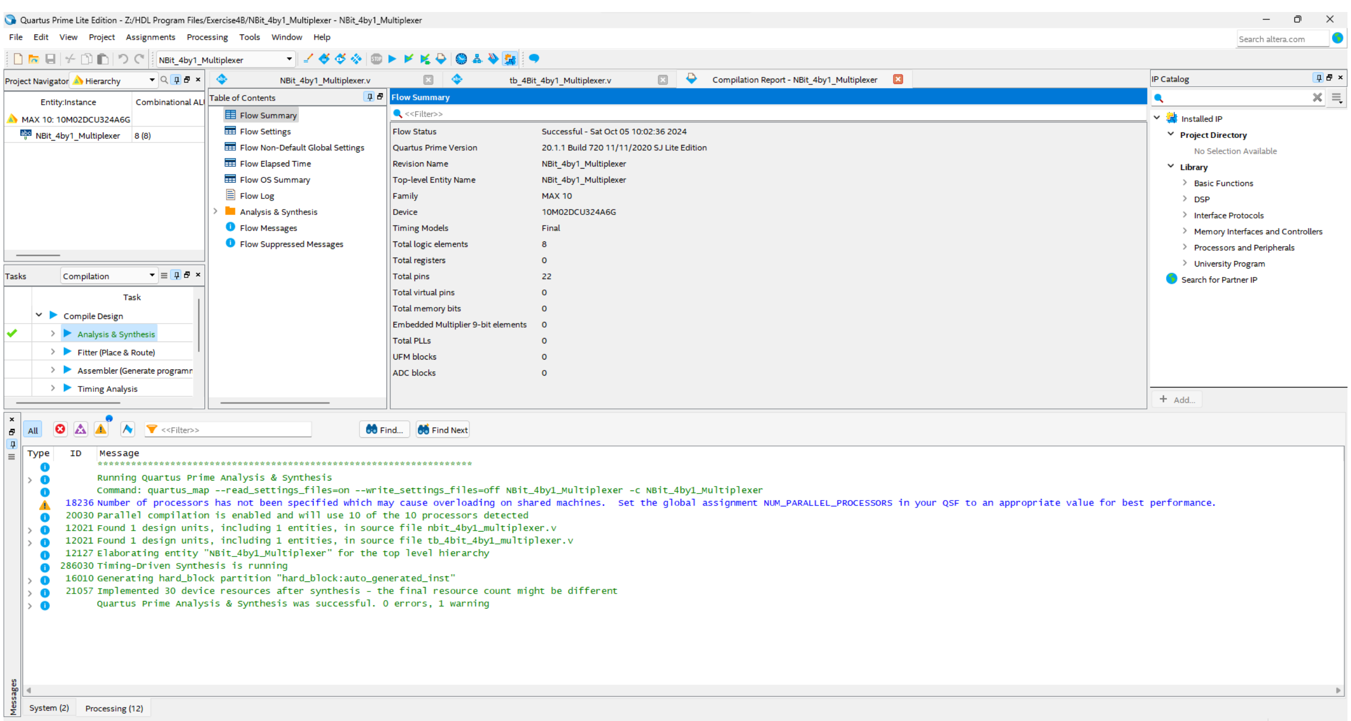
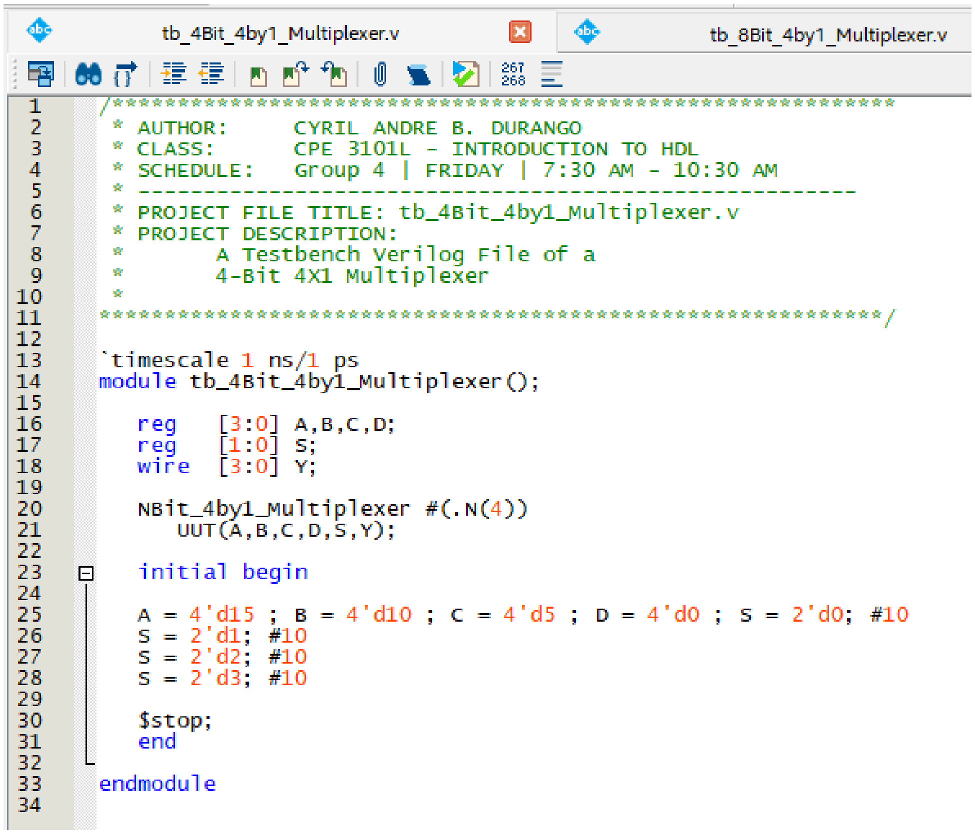


Figure 6. Verified Summary Compilation for the N-bit 4-by-1 Multiplexer



```verilog
/*************************************************************
 * AUTHOR:     CYRIL ANDRE B. DURANGO
 * CLASS:      CPE 3101L - INTRODUCTION TO HDL
 * SCHEDULE:   Group 4 | FRIDAY | 7:30 AM - 10:30 AM
 * -------------------------------------------------------
 * PROJECT FILE TITLE: tb_4Bit_4by1_Multiplexer.v
 * PROJECT DESCRIPTION:
 *       A Testbench Verilog File of a
 *       4-Bit 4X1 Multiplexer
 *
*************************************************************/

`timescale 1 ns/1 ps
module tb_4Bit_4by1_Multiplexer();

   reg   [3:0] A,B,C,D;
   reg   [1:0] S;
   wire  [3:0] Y;

   NBit_4by1_Multiplexer #(.N(4))
      UUT(A,B,C,D,S,Y);

   initial begin

   A = 4'd15 ; B = 4'd10 ; C = 4'd5 ; D = 4'd0 ; S = 2'd0; #10
   S = 2'd1; #10
   S = 2'd2; #10
   S = 2'd3; #10

   $stop;
   end

endmodule
```

Figure 7. Testbench File of a 4-bit 4-by-1 Multiplexer Circuit

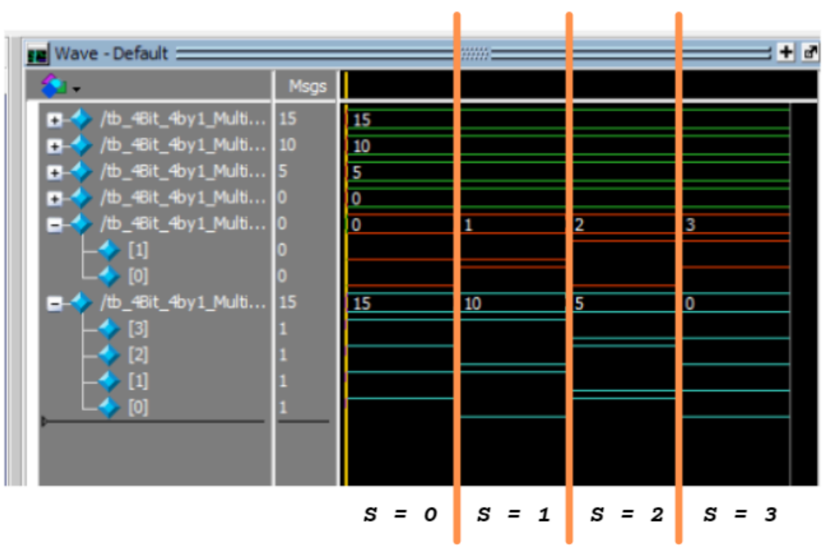


Figure 8. Annotated Generated Waveform of a 4-bit 4-by-1 Multiplexer Circuit

The above waveform shows the selector characteristics of the multiplexer. When S = 0, the output waveform generated is the binary equivalent of what is inputted in the 0th location. The same logic applies to the rest of the inputs and its corresponding selctor.

tb_8Bit_4by1_Multiplexer.v | NBit_4by1_Multiplexer.v

```verilog
/*****************************************************************
 * AUTHOR:      CYRIL ANDRE B. DURANGO
 * CLASS:       CPE 3101L - INTRODUCTION TO HDL
 * SCHEDULE:    Group 4 | FRIDAY | 7:30 AM - 10:30 AM
 * ---------------------------------------------------------
 * PROJECT FILE TITLE: tb_8Bit_4by1_Multiplexer.v
 * PROJECT DESCRIPTION:
 *         A Testbench Verilog File of a
 *         8-Bit 4X1 Multiplexer
 *
*****************************************************************/

`timescale 1 ns/1 ps
module tb_8Bit_4by1_Multiplexer();

   reg    [7:0] A,B,C,D;
   reg    [1:0] S;
   wire   [7:0] Y;

   NBit_4by1_Multiplexer #(.N(8))
      UUT(A,B,C,D,S,Y);

   initial begin

   A = 8'd255 ; B = 8'd128 ; C = 8'd200 ; D = 8'd171 ; S = 2'd0; #10
   S = 2'd1; #10
   S = 2'd2; #10
   S = 2'd3; #10

   $stop;
   end

endmodule
```

Figure 9. Testbench File of an 8-bit 4-by-1 Multiplexer Circuit

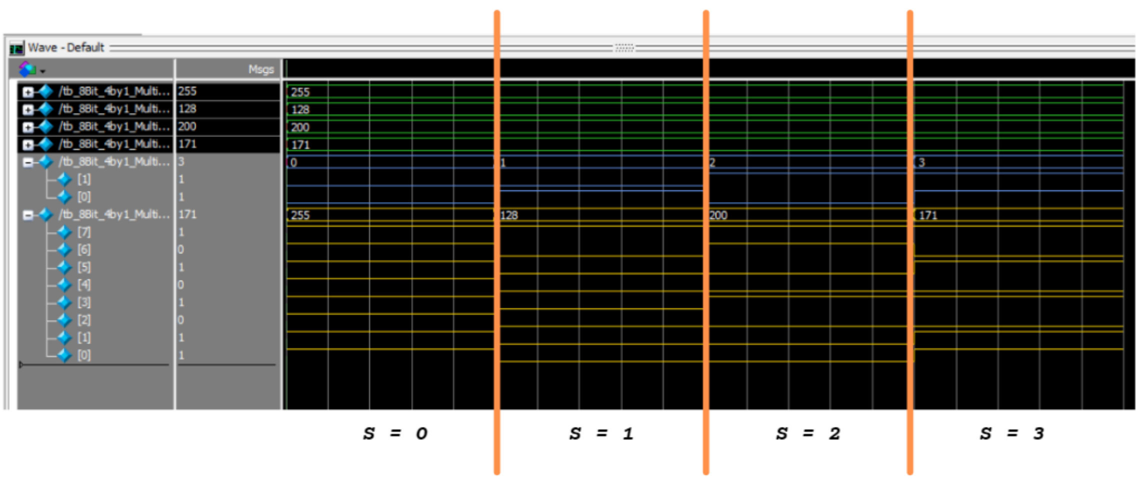


Figure 10. Annotated Generated Waveform of a 4-bit 4-by-1 Multiplexer Circuit

The above waveform shows the selector characteristics of the multiplexer. When S = 0, the output waveform generated is the binary equivalent of what is inputted in the 0th location. The same logic applies to the rest of the inputs and its corresponding selctor.

It can also be observed in both generated waveforms for the 4-bit and 8-bit versions of the 4-by-1 Multiplexer Circuit, the input groups are always triggered. It will only be outputted by the time the selector is configured to its corresponding input group. A is outputted when S = 0, B is outputted when S = 1, C is outputted when S = 2, and D is outputted when S = 3